Deliverable #3 : Detailed Design

SE 3A04: Software Design III – Large System Design

Tutorial Number: T02
Group Number: G5
Group Members:

- Anura, Mithu
- Banik, Tridib
- Khorramijam, Arsia
- Pham, Aaron
- Zhang, Han

1 Introduction

1.1 Purpose

This document provides further information about the SCEMAS system architecture, including state chart diagrams, sequence diagrams, and a detailed class diagram.

This document is intended for internal SCEMAS stakeholders, including but not limited to project managers, developers, domain experts, and investors. SCEMAS Deliverables 1 and 2 should be read prior, and technical knowledge may be beneficial in better understanding the contents of the document.

1.2 System Description

An overview of the high-level architectural design can be found in deliverable 2. This document acts as an extension of deliverable 2, providing more context in the form of state charts, sequence diagrams, and a detailed class diagram.

1.3 Overview

This document is organized by chart/diagram type. Section 2 contains relevant state chart diagrams, Section 3 contains relevant sequence diagrams, and Section 4 provides a detailed UML class diagram of the system.

2 State Charts for Controller Classes

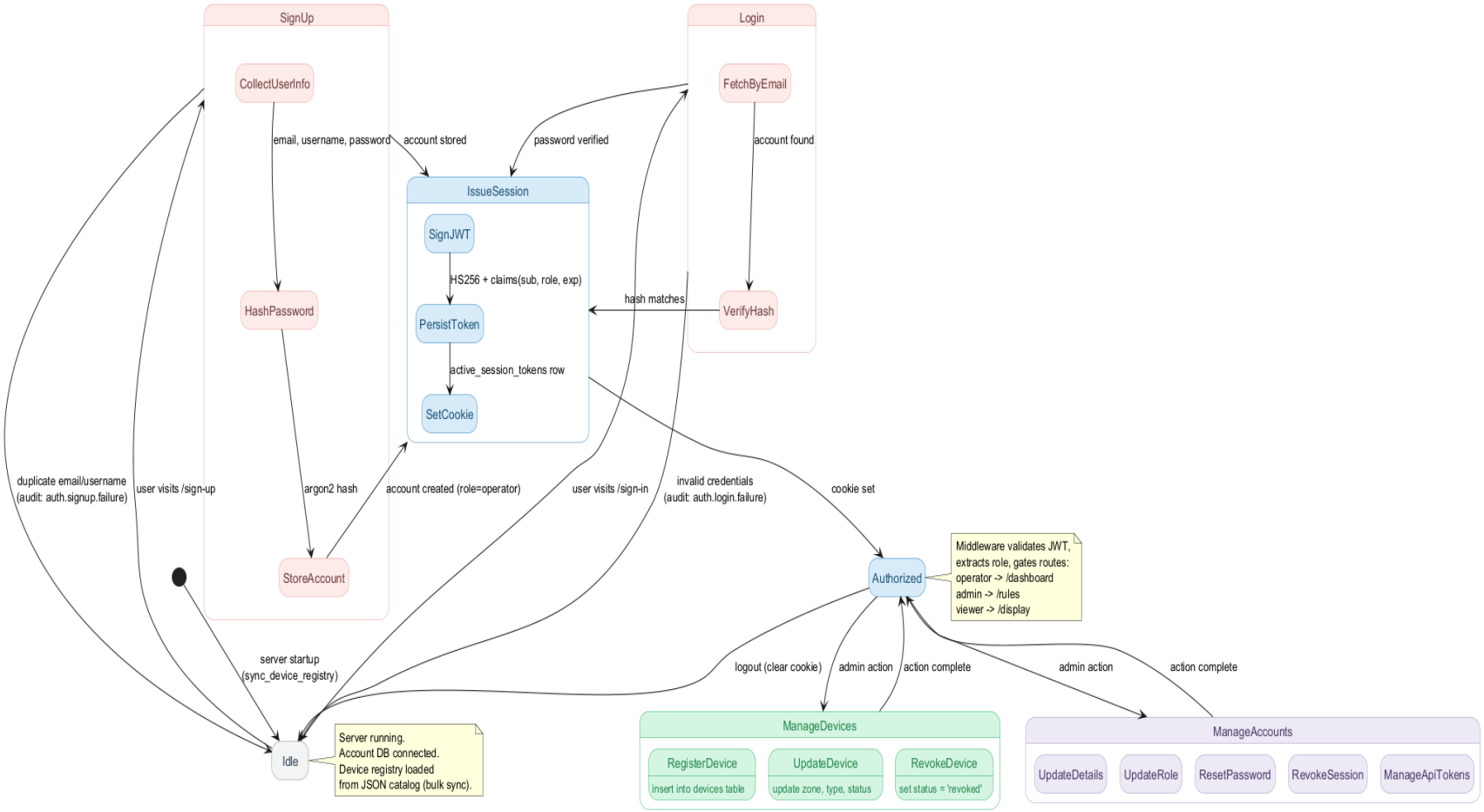


Figure 2.1: Overall Access Management Controller State Chart Diagram

The enlarged overall Access Management State Chart Diagram is shown in Figure 2.1.1 and Figure 2.1.2

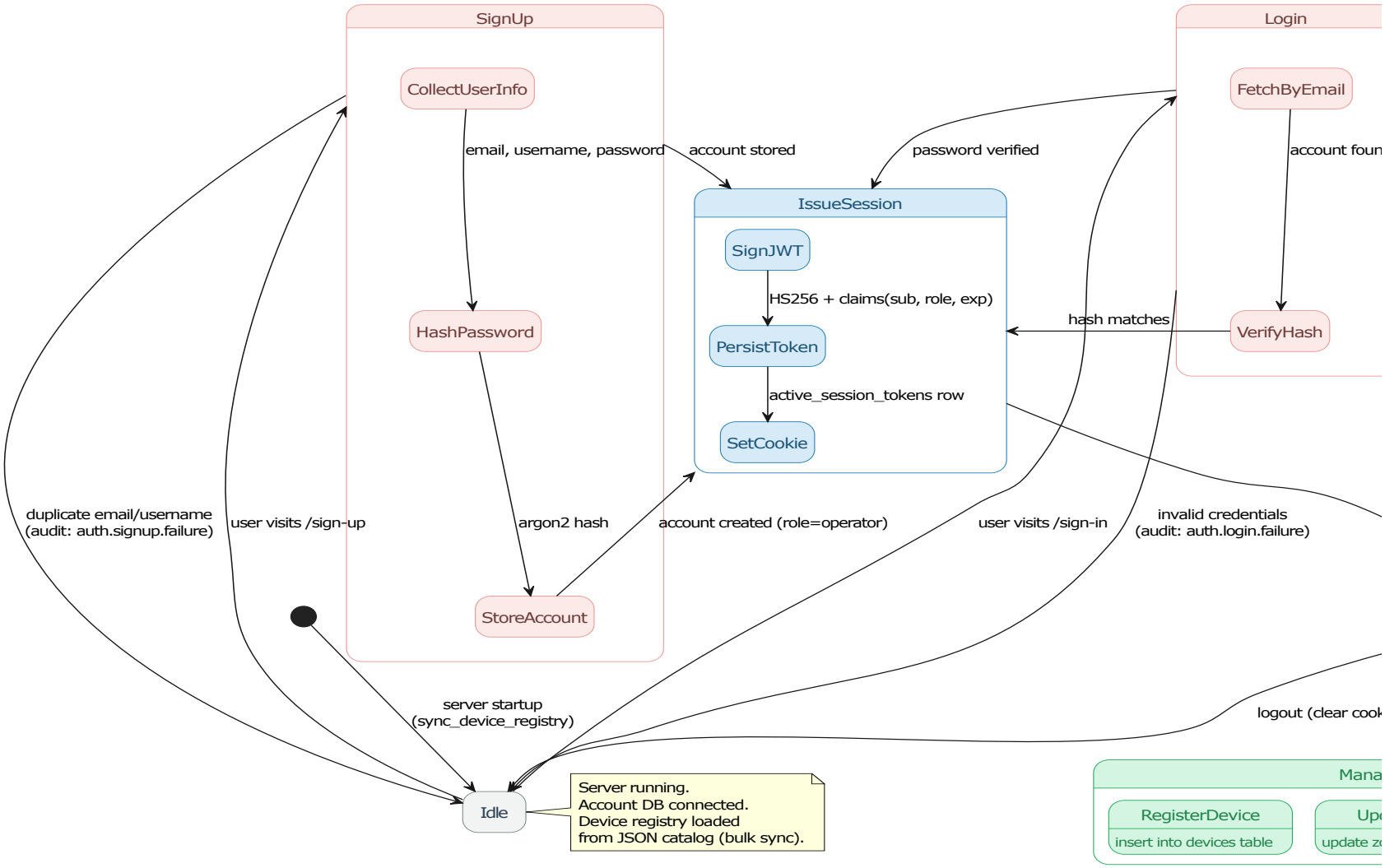


Figure 2.1.1: Access Management Controller State Chart Diagram (Part 1)

Figure 2.1.1 is on the left of Figure 2.1.2

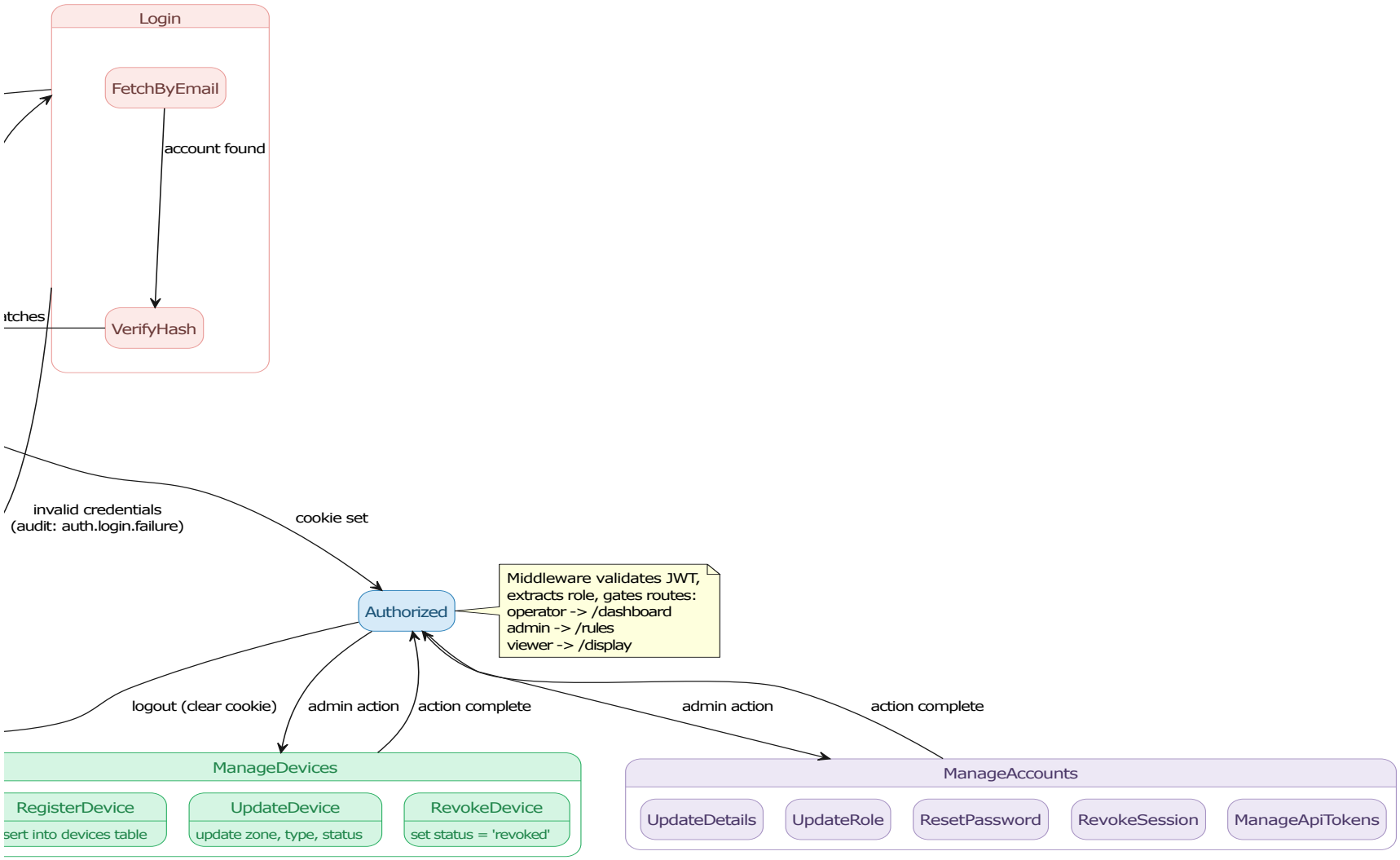


Figure 2.1.2: Access Management Controller State Chart Diagram (Part 2)

Figure 2.1.2 is on the right of Figure 2.1.1

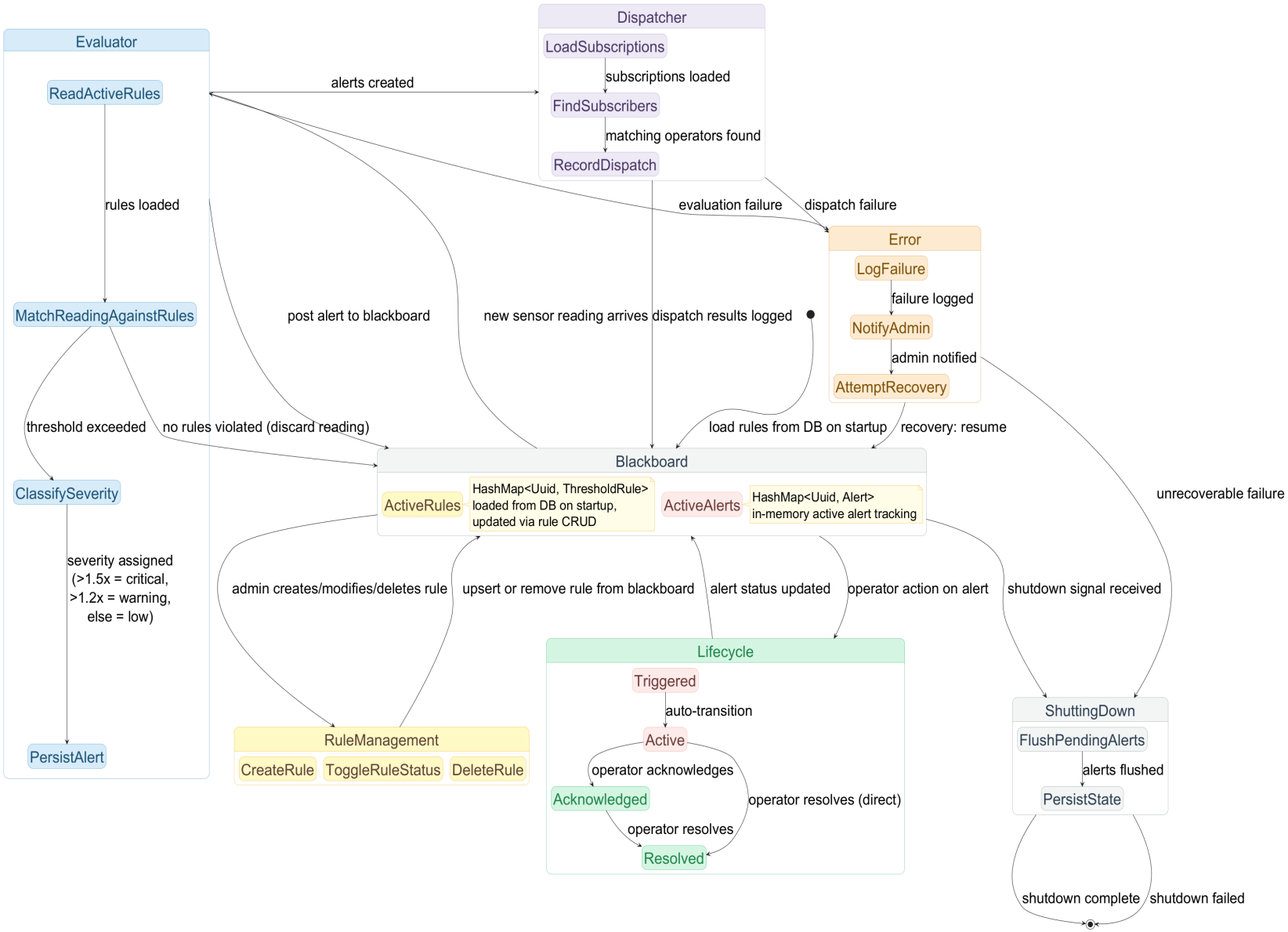


Figure 2.2: Alerting Management Controller State Chart Diagram

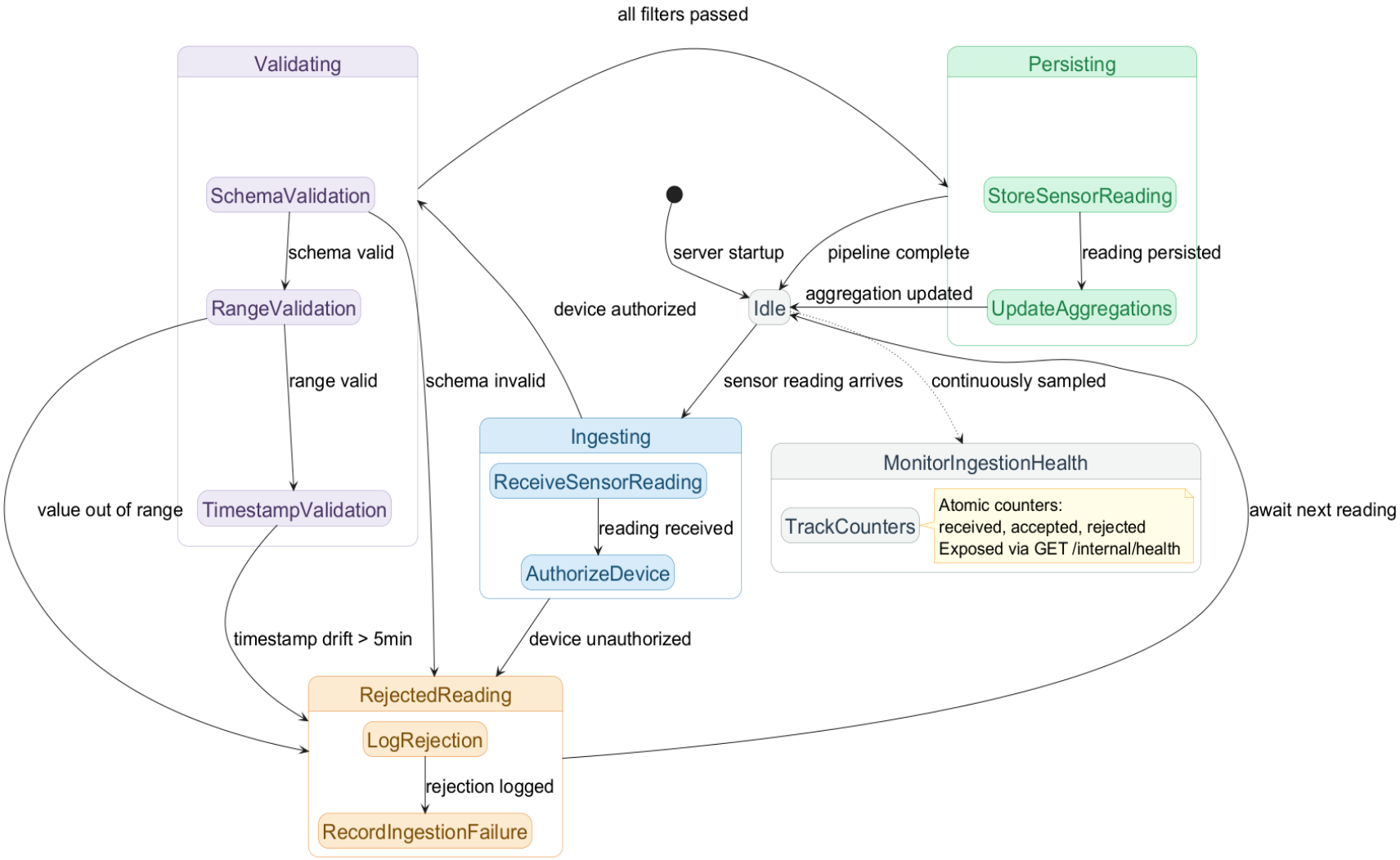


Figure 2.3: Telemetry Management Controller State Chart Diagram

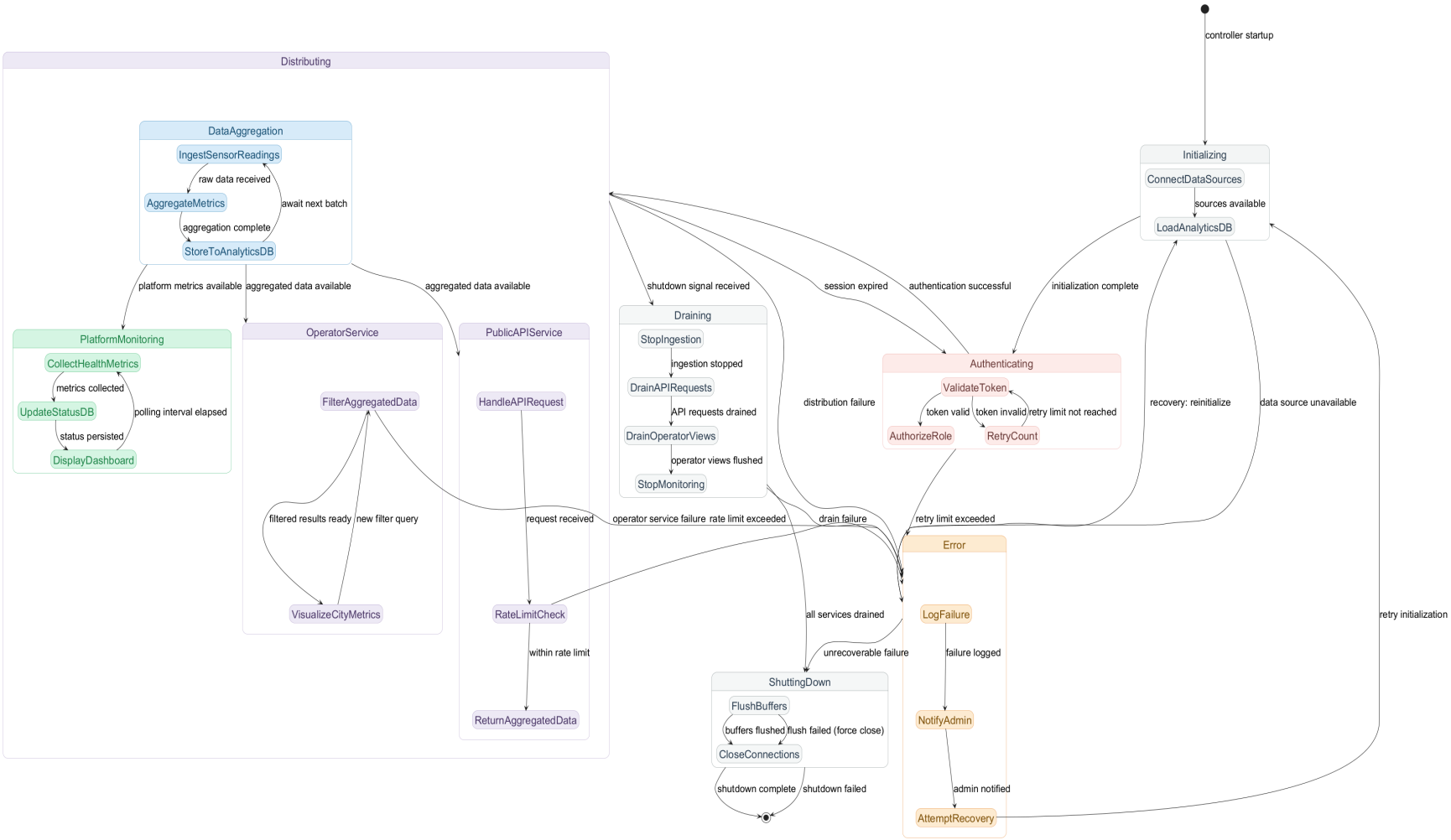


Figure 2.4: Data Distribution Management Controller State Chart Diagram

The enlarged overall Data Distribution Management Controller State Chart Diagram is shown in Figure 2.4.1 and Figure 2.4.2

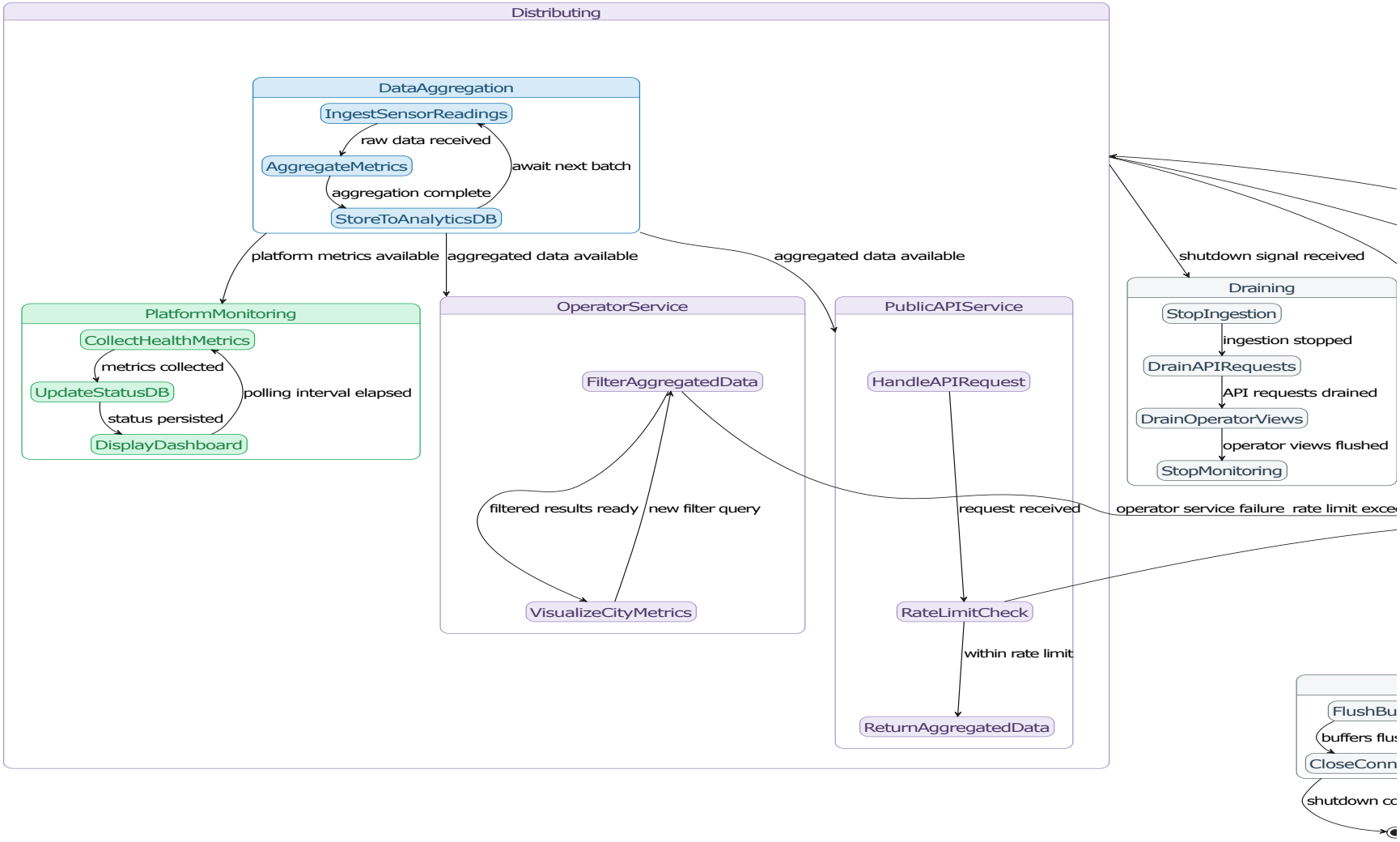


Figure 2.4.1: Data Distribution Management Controller State Chart Diagram (Part 1)

Figure 2.4.1 is on the left of Figure 2.4.2

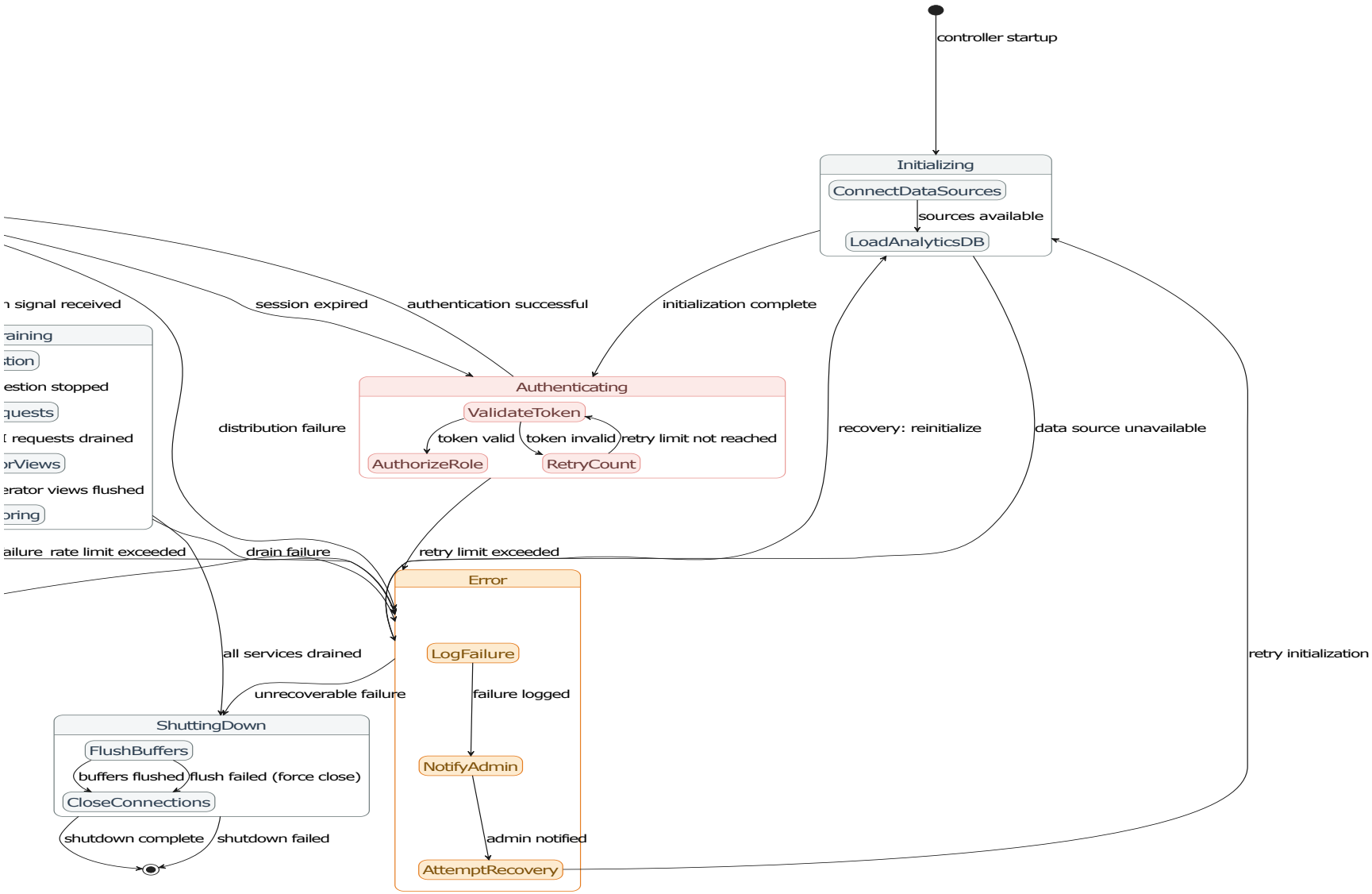


Figure 2.4.2: Data Distribution Management Controller State Chart Diagram (Part 2)

Figure 2.4.2 is on the right of Figure 2.4.1

3 Sequence Diagrams

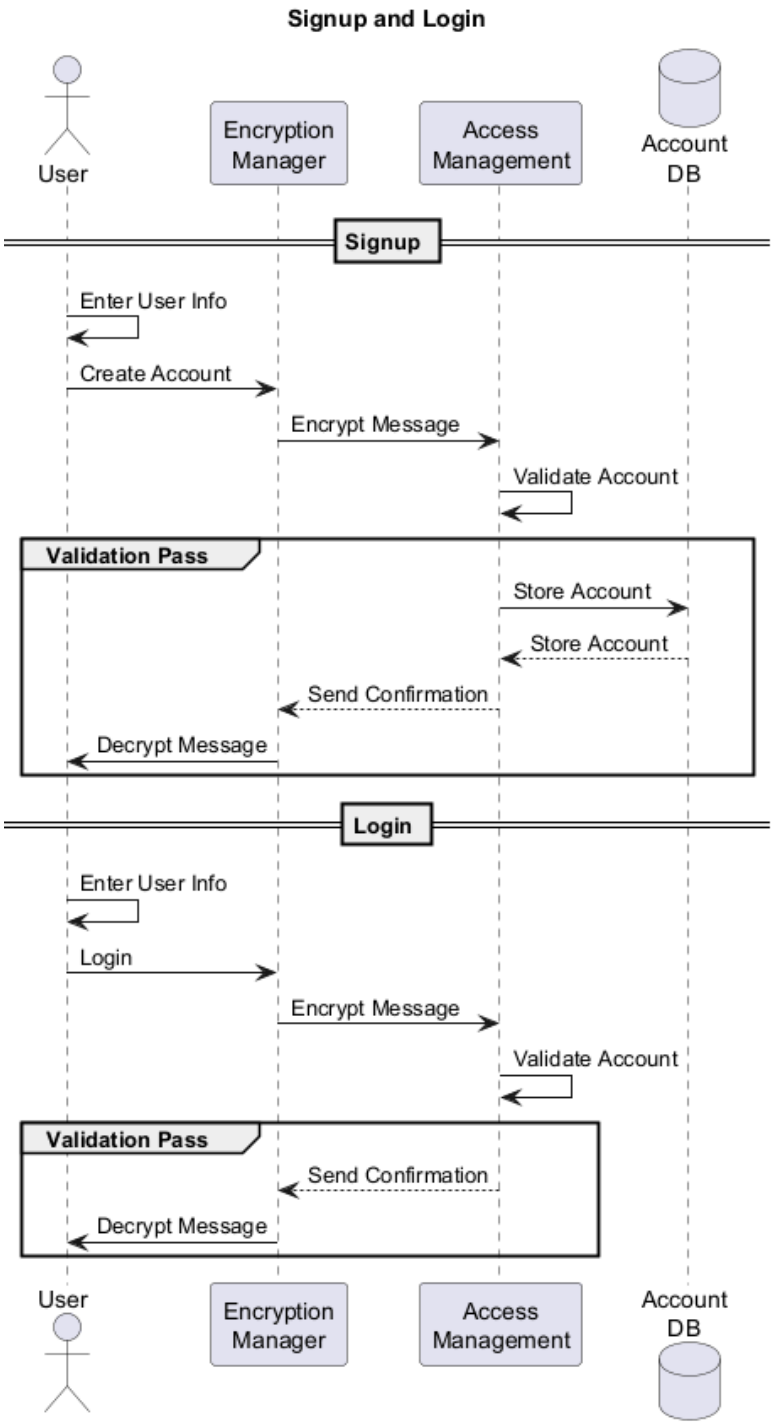


Figure 3.1: Sign Up and Login Sequence Diagram

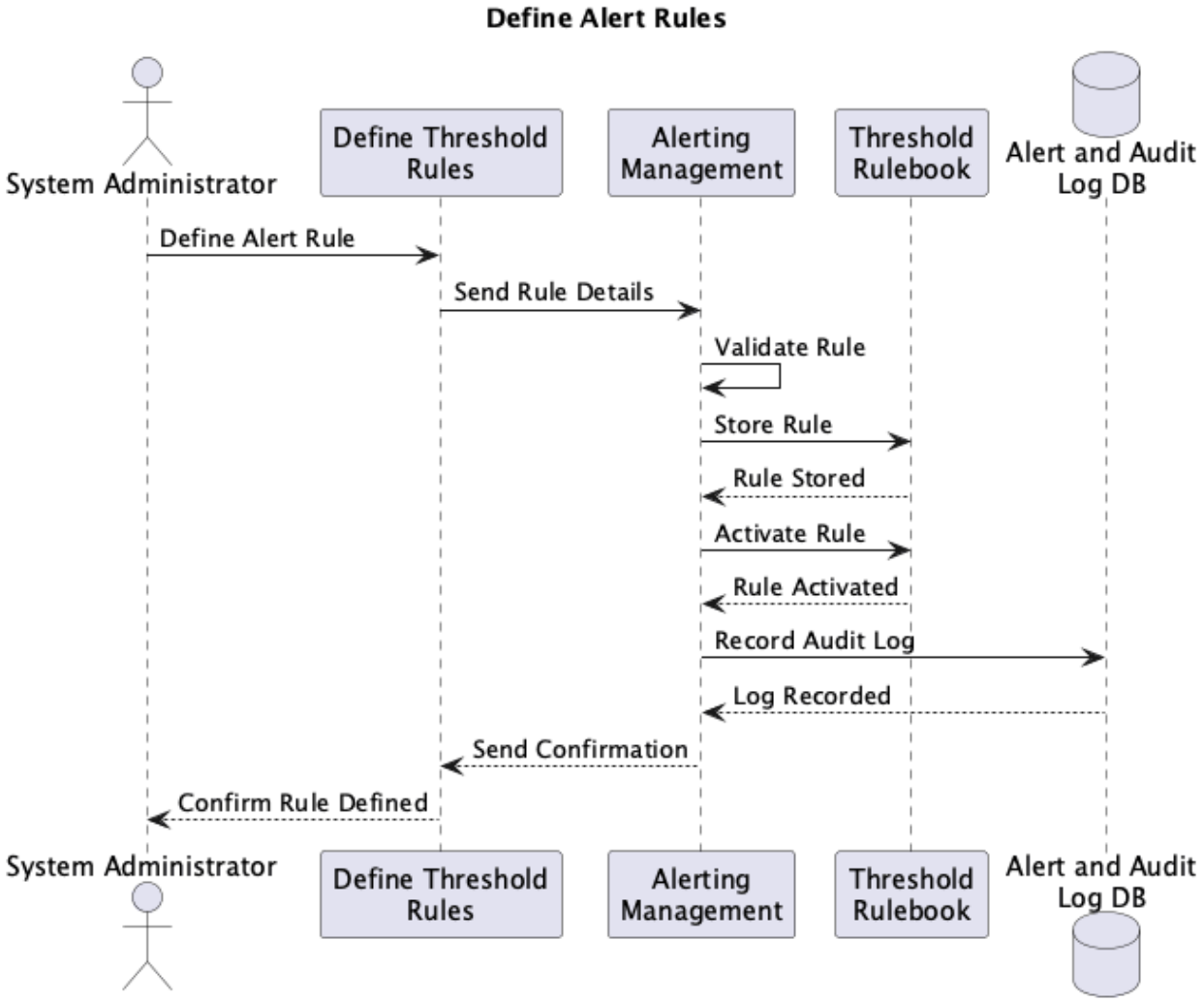


Figure 3.2: Define Alert Rule Sequence Diagram

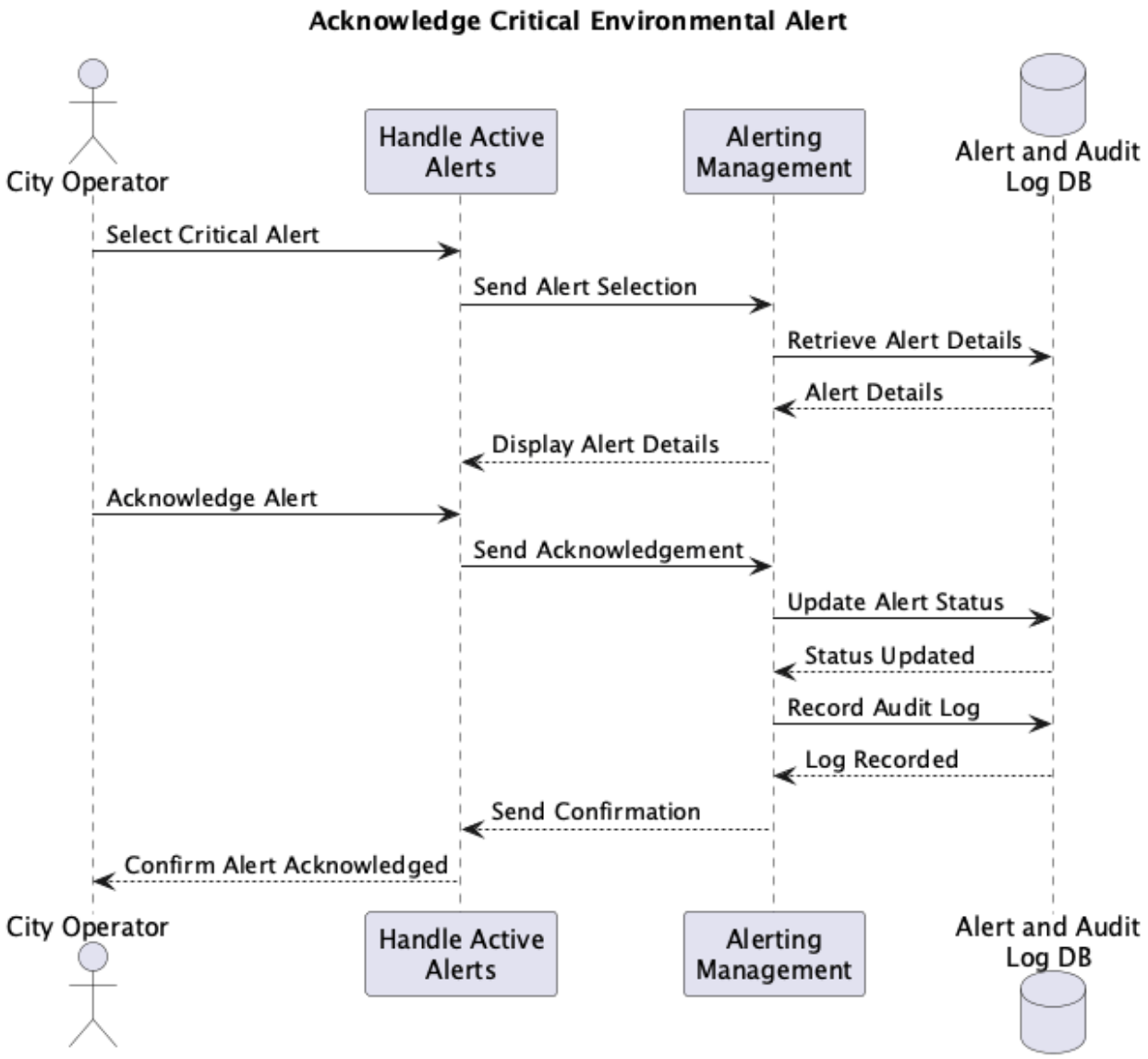


Figure 3.3: Acknowledge Critical Environment Alert Sequence Diagram

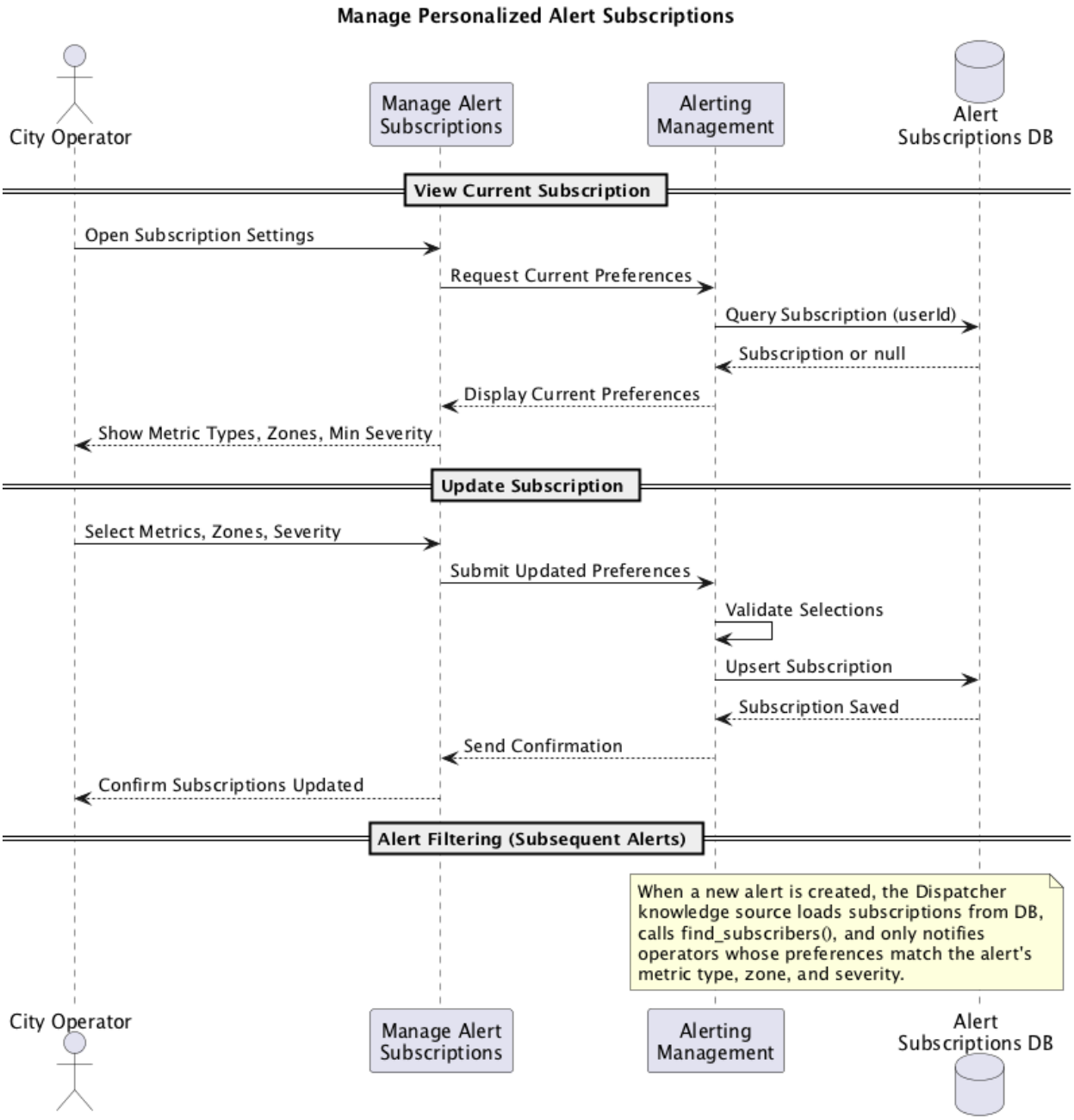


Figure 3.4: Manage Alert Subscriptions Sequence Diagram

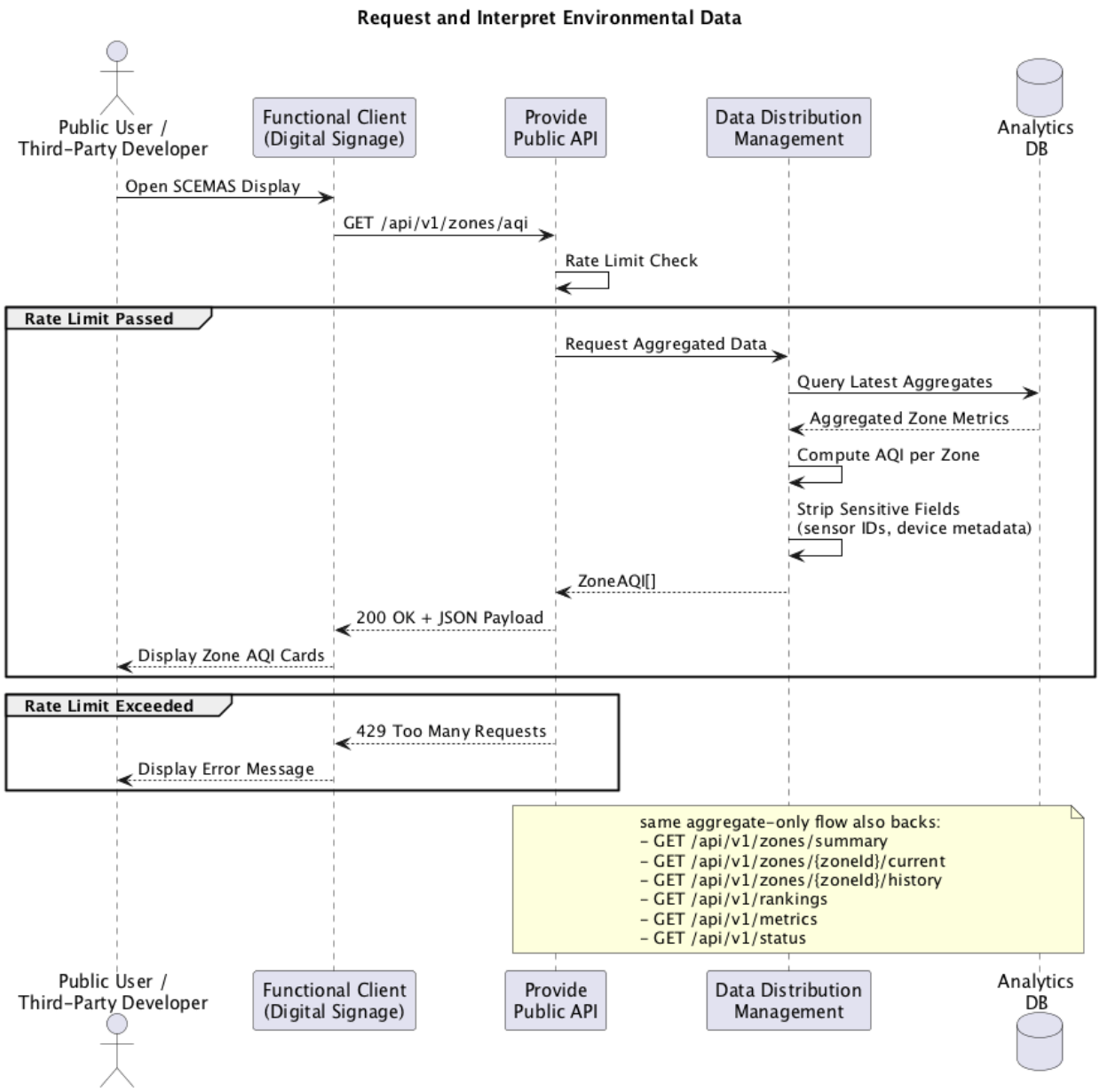


Figure 3.5: Request and Interpret Data Sequence Diagram

4 Detailed Class Diagram

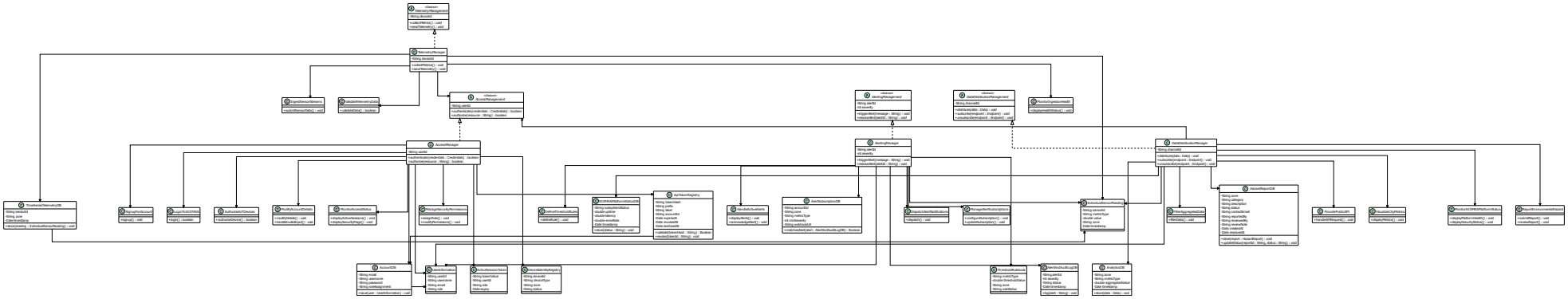


Figure 4.1: Overall Detailed Class Diagram

The enlarged Overall Detailed Class Diagram is shown in Figure 4.1.1, Figure 4.1.2, and Figure 4.1.3

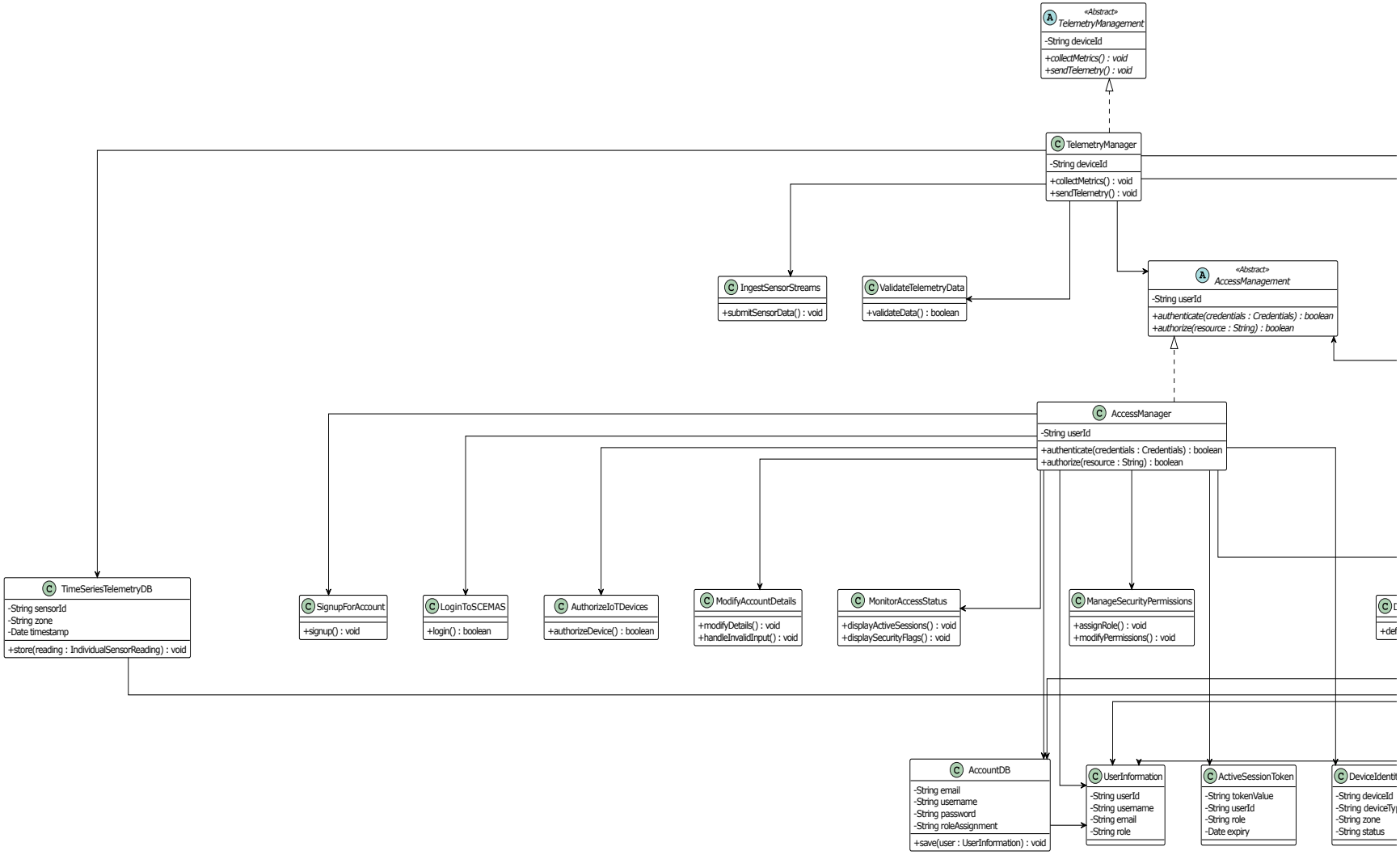


Figure 4.1.1: Detailed Class Diagram – Part 1: Telemetry & Access

Figure 4.1.1 is on the left of Figure 4.1.2

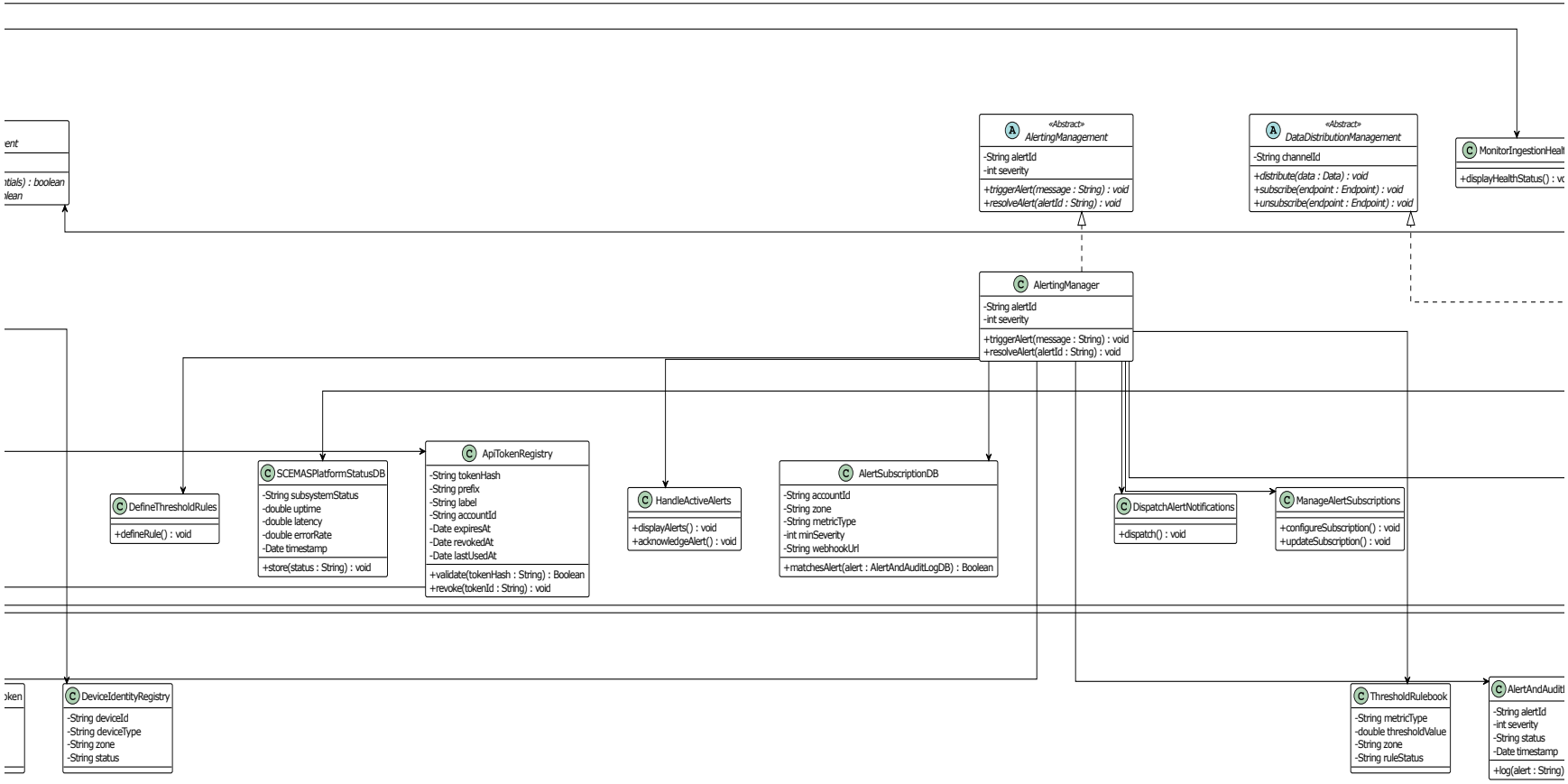


Figure 4.1.2: Detailed Class Diagram – Part 2: Alerting

Figure 4.1.2 is on the left of Figure 4.1.3

Figure 4.1.2 is on the right of Figure 4.1.1

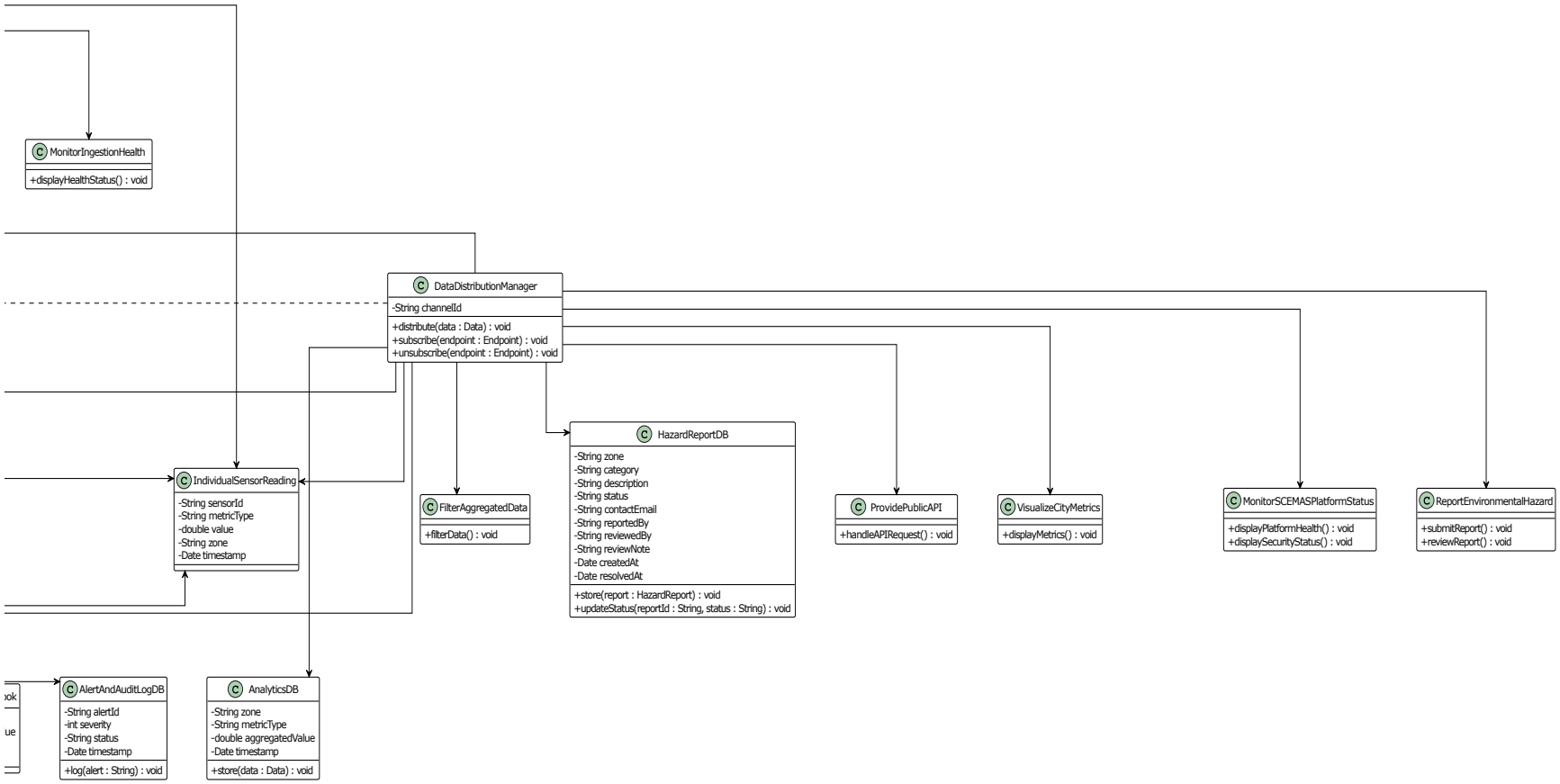


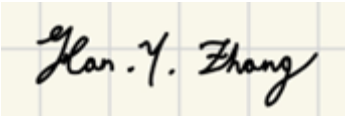
Figure 4.1.3: Detailed Class Diagram – Part 3: Data Distribution

Figure 4.1.3 is on the right of Figure 4.1.2

A Division of Labour

Zhang, Han:

- Access Management Controller State Chart Diagram.
- Request and Interpret Environmental Data Sequence Diagram.



Pham, Aaron:

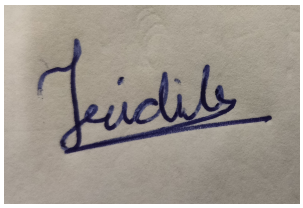
- Data Distribution Management Controller State Chart Diagram.
- Manage Alert Subscriptions Sequence Diagram.
- Created the final version of Detailed Class Diagram, added the classes related to innovative features; i.e., AlertSubscriptionDB and updated the relationships related to the new classes.

Anura, Mithu:

- Alerting Management Controller State Chart Diagram.
- Telemetry Management Controller State Chart Diagram.

Khorramijam, Arsia:

- Define Alert Rules Sequence Diagram.
- Acknowledge a Critical Environmental Alert Sequence Diagram.

Banik, Tridib:

- Section 1: Introduction.
- Sign Up and Login Sequence Diagram.
- Created the initial draft of the Detailed Class Diagram based on CRC cards from deliverable 2.
- Added all diagrams to LaTeX, added figure captions, adjusted figure sizes for readability.